Engineering Mechanics 1 · Unit_01 · UNIT 1

DRAFT — UE26ME142A · Unit_01 · UNIT 1 · UE26ME142A Unit 1 Class 6 AVSummary

Standalone video lesson · authored study notes with source citations

Hugging Face PESU_Materials source tree, official units.json catalogue, transcript sidecars, slide text/PDF/PPT sources and the selected First_Year_8enterprise archive; see the provenance datasets in the release ZIP. Generated 18 October 2026.

· How to use this pack

Read the topic explanation, worked examples, and recap. Evidence citations point to source transcript intervals and slide pages; full slide images remain in the source visual index. Topics include the applicable parts: overview, key terms, formulas, figures, flowcharts, important points, worked examples, past-paper questions and a recap. IDs are shared with the separate documents: E = formula, G = graph, F = flowchart, W = worked example. Each ID is numbered by topic, so E4.2 is the second formula of topic 4.

Source text	Verbatim or OCR-derived text, kept separate from editorial notes.
Candidate	Automated topic/source association requiring manual confirmation.
Low ASR confidence	Transcript segment requires audio verification.
DRAFT	Human review remains incomplete.

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M1

UE26ME142A Unit 1 Class 6 AVSummary

1 UE26ME142A Unit 1 Class 6 AVSummary

Study focus: UE26ME142A_Unit 1_Class 6_AVSummary. This topic explains the central idea, its conditions, and how the lecture applies it in context. Review the selected source statements and follow their citations for the full lesson segment. EV0000146 · EV0000147

SHORT-NOTE RECAP

Core explanation · 00:00–04:59

My name is Dr.

Key details

- Swee Venkadesh from Civil Engineering under Department of Science and Humanities, PS University.
- In the last class we discussed moment and solved a few problems on movement.
- That is in class number eight.
- This is class number nine.
- We continue to solve a few more problems on moment.
- The statement of the problem is in order to raise the flat pole, OC.
- A light frame OAB is attached to the pole and a tension of 3.2 kN is developed in the hosting cable by the power range D.
- Calculate the moment M naught this should be M naught suffix not M naught of this tension about the h point O.
- From the statement, what is that we should think or we should get is the tension in the cable, that is this cable, is 3.2.
- It is tension, so external forces will be moving away, the internal forces will be moving towards that means the force in this cable internal force will be from A towards D and at D it is being from D towards A.
- What would be the movement caused by this force applied onto this light frame AOB at point O because this flat pole when I apply this tension here.
- This has to rotate and stand vertical.
- That's the whole idea of this mechanism.
- Force which is 3.2 kilonewton.

- What are we supposed to find is the moment that oh because of this force that is M naught.
- We are supposed to identify that this OA line is inclined at twenty degrees.
- Similarly, if I draw an horizontal line here passing through A, this angle will also be equal to 20 degrees because these are alternate angles, they will be equal.
- That means it forms an isoscelus triangle.
- In case of an equilateral triangle, the internal angle of the triangle, all of them will be equal.
- That means it will be equal to 180 divided by 3.
- Teta if I take T is the angle made by the internal angle of the triangle that is 180 by 3, which would be 60 degrees.
- For this force, what I need to do is I should take this force, resolve the force along horizontal and along vertical, determine what is the vertical distance from O to A.
- Sine of this angle which is sixty degrees.
- To find out the component of the force, horizontal component, it is 3.2, this angle will be 20, so cos 20, whereas the vertical component will be 3.2 sin 20.
- into the horizontal distance between OA, which is 3 into cos 20, which will produce a clockwise movement, so that's why it is negative.
- So that would be the moment at point O because of the application of the force equal to 3.2 RTA, which is the tensile form.
- Next problem is 2 bar 56.
- The mechanism shown is used to lower disabled persons into a very ball tube or therapeutic.

Example or application from the lecture

- Then the sides of this triangle OAB is given as 3 meters.

These are edited study notes based on the cited lecture interval. Check uncertain speech recognition against the source transcript and recording.

EV0000146 · EV0000147 · EV0000148 · +1 source units in pack.json

Core explanation · 05:00–09:59

Treatment in the uploaded configuration, the weight of the boom and hanging chair induces a compressive force of 5 sub 25 newton in hydraulic cylinder A B.

Key details

- Compressive means that the force which cylinder AB exerts on point B is directed from A towards B.
- If theta is equal to 30 degrees, determine the movement of this cylinder force acting on pin B about A point O and B point C.
- Which is used for lowering the disabled person into the will pole.
- Which is a com which is going to be a compressive force.
- He says that if it is compressive, means that the force is exerted from.
- If I from A to B at point B that is this is the force that is the internal force.
- The compressive force means the external forces will be moving towards each other, the internal forces will be moving away.
- We have to find out moment about O and moment about C.

- We can make out the distances from O to A, the vertical distance from O to A, which is 200, from A to C as 800, from C to this point as 175, from this point to this point as 100.
- Then the inclined distance along this point to this point is 400, then from this point to this edge is 900, then very important is this distance from B to this.
- We have a force acting at B.
- First thing is if I draw a vertical line from B till the horizontal line extended from A, it will cut at some point here.
- Now first thing is to find out this angle.
- Why should we find out this angle?
- To find this angle,.
- I should get this what horizontal distance, sorry, vertical distance and then horizontal distance.
- Vertical distance I can find it as 800 plus 175 plus 100 plus this small distance which I can take it as distance from this point to here.
- Which I'll get it from this 400, from this 400 sine of theta, I'll get this angle, this distance, sorry, this distance.
- Then I'll subtract from 150 this because this is 90 degrees.
- The horizontal distance between this also I can determine 400 cos theta I'll get this distance from here to here.
- Or find out 5, if I want to find out 5, which will be equal to 10 inverse of this vertical distance and this horizontal distance.
- What would be this vertical distance?
- 800 as I have told you explained, 800 plus 175 plus this is 400.

Example or application from the lecture

- The magnitude of the force is given as 575 newtons.
- That will be the sense of the force, and he also says if the angle theta that is this angle, the boom angle with respect to the horizontal is 30 degrees, what is given here, then determine what is the movement caused by this force about point O and about point A.

These are edited study notes based on the cited lecture interval. Check uncertain speech recognition against the source transcript and recording.

EV0000150 · EV0000151 · EV0000152 · +1 source units in pack.json

Core explanation · 10:00–14:59

If I take sine theta, I get this vertical distance, but I have to subtract this vertical distance from this to this.

Key details

- Cause of this angle theta, this would be theta itself because this line and this line are not 90 degrees.
- If this is theta, if I draw a line here, okay, if this is theta, this will be 90 minus theta and this would be theta.
- That is this distance, 175 plus 400, which is this, cause of this theta, I get it till this point, so 175 plus this, plus I should take this horizontal distance, which is 150 sin theta, but I will be

getting from here to this point.
- Once I get this phi, I can resolve this force as 575 cos of phi, I get the.
- Vertical force 575 sine of 5, I get the horizontal force.
- The vertical distance would be this, plus I need to add this 200, or in another way, I can take the point of application of this force as A itself.
- Then this horizontal component is to 200, which will be clockwise.
- I'll take this vertical component into this distance 200, this horizontal component into this distance which is 800.
- 575 sin 5 into 0.2 minus 575 cos 5 into 0.2 we will get the moment about O M naught.
- Moment about C will be 575 sin 5 into 0.2 plus 575 cos 5 into 0.8.
- I get this moment.
- I hope you have understood this.
- Next problem is 5 bar or sorry 2 bar 557.
- The asymmetrical support arrangement is chosen for a pedestrian bridge because conditions at the right end F.
- Do not permit a support tower and anchorage.
- During a test, the tensions in cable 2, 3, and 4 are all adjusted to the same value T.
- If the combined movement of all four cable tensions.
- About point O is to be 0.
- What should be the value T1 of the tension in cable 1?
- That is what is asked.
- Then determine the corresponding value of the compressive force P.
- At O resulting from the four tensions applied at A.
- Neglect the weight of the tower.
- Support, tower, or anchorage.
- Support reaction at B F is zero.
- There is no support here.
- All the supports has to come through the cables.
- The tension in cable two that is this ten, this cable, then three this cable, four that is this cable, all of them have the same tension T, the same tension T in this two, three and four cable.

Example or application from the lecture

- I should not consider this minus 200, only this part, the total.

These are edited study notes based on the cited lecture interval. Check uncertain speech recognition against the source transcript and recording.

EV0000154 · EV0000155 · EV0000156 · +1 source units in pack.json

Core explanation · 15:00–19:59

Now for this condition we have to find out what would be the tension in this cable AB, that is cable 1.

Key details

- Then we should also find out what would be the compressive force in this tower AO acting at O.
- We have to neglect the weight of this tower.
- What has to be found also I have explained.
- Next, we should also find out what is the tension in P1 and 10 and the compressive force that is acting on the tower P.
- We should take T as a known quantity that is T.
- Intention T2, T3, and T4 is equal to T.
- Now to find out the movement of O, because he says that the combined movement of all the four forces at point O is 0.
- Then this is an horizontal line.
- Note that the.
- This is not vertical.
- OA is not vertical, it is inclined to vertical at an angle of 5 degrees.
- Theta 2 will be this, which will be equal to.
- The vertical distance which is 15 into the horizontal distance 9 or divided by 9.
- Corresponding to the inclination of the cable 1, 2, 3, and 4 with the horizontal.
- Then T1 is the tension acting which I will take it at D.
- I will resolve all these forces along vertical and horizontal.
- I can find out which will be from this distance and the distance between this.
- I can find out that.
- And that is what is explained here.
- So I've taken T out because that is common.
- Minus T1 cos teta 1.
- into the vertical distance of all of them are 4.8 itself.
- This is the vertical distance.
- All of them will have the same vertical distance 4.8.
- Into I should take this distance from O to B, which will be 6.6.
- That is from the O to A to B, the horizontal distance, then I should also take what is the horizontal distance from A to O, which will be equal to 15 plus 4.8, which is 19.8.

Example or application from the lecture

- The horizontal distance is given as 6.6.
- The vertical distance is given as 15.
- Tower AO is inclined at an angle of 5 degrees, which is given here.

These are edited study notes based on the cited lecture interval. Check uncertain speech recognition against the source transcript and recording.

EV0000158 · EV0000159 · EV0000160 · +1 source units in pack.json

Core explanation · 20:00–24:59

And this angle is given us 5 degrees.

Key details

- That is what is entered here.

- Similarly, I will find out for all these three vertical forces acting because of these tensions and the distances.
- But here I have to subtract the horizontal distance from O to A.
- If T is in Newton or whichever units that T is going to be.
- T is the tension in the cable, 2, 3, and 4.
- Next to find out the compressive force in OA, I will take some of all the forces acting along the vertical.
- Note that the.
- Compressive force that is acting in OA will be acting upwards, but that is at inclined at 5 degrees to the vertical.
- That should be equal to all these forces which is acting.
- P1 and this, all this which should be acting downwards.
- Attention in the table two, three, and four.
- I hope all of you have understood today's class.
- We will continue in the next class.
- This is class number nine.
- And I am Dr.
- SV Venkatesh from PS University.
- We will continue in the next class.

These are edited study notes based on the cited lecture interval. Check uncertain speech recognition against the source transcript and recording.

EV0000162 · EV0000163 · EV0000164 · +1 source units in pack.json

A Symbol glossary

Symbol	Name	Meaning	Appears in

B Sources, coverage and gaps

Source	Path	Use
UE26ME142A_Unit 1_Class 6_AVSummary transcript (JSON)	https://huggingface.co/buckets/alvin3y10/data/tree/pesuc/PESU_Materials/Sem-1/UE26ME142A_Engineering_Mechanics_1/Unit_01/AV_Summaries/006_UE26ME142A_Unit_1_Class_6_AVSummary__f35d743d3b761113.transcript.json	Full JSON sidecar. Normalized text similarity to canonical JSON: 1.0000; only the JSON segmentation supplies the displayed timestamps.

Source	Path	Use
UE26ME142A_Unit 1_Class 6_AVSummary transcript (Markdown)	https://huggingface.co/buckets/alvin3y10/data/tree/pesuc/PESU_Materials/Sem-1/UE26ME142A_Engineering_Mechanics_1/Unit_01/AV_Summaries/006_UE26ME142A_Unit_1_Class_6_AVSummary__f35d743d3b761113.transcript.md	Full Markdown sidecar. Normalized text similarity to canonical JSON: 1.0000; only the JSON segmentation supplies the displayed timestamps.
UE26ME142A_Unit 1_Class 6_AVSummary transcript (plain text)	https://huggingface.co/buckets/alvin3y10/data/tree/pesuc/PESU_Materials/Sem-1/UE26ME142A_Engineering_Mechanics_1/Unit_01/AV_Summaries/006_UE26ME142A_Unit_1_Class_6_AVSummary__f35d743d3b761113.transcript.txt	Full plain text sidecar. Normalized text similarity to canonical JSON: 0.9995; only the JSON segmentation supplies the displayed timestamps.
UE26ME142A_Unit 1_Class 6_AVSummary transcript (SRT)	https://huggingface.co/buckets/alvin3y10/data/tree/pesuc/PESU_Materials/Sem-1/UE26ME142A_Engineering_Mechanics_1/Unit_01/AV_Summaries/006_UE26ME142A_Unit_1_Class_6_AVSummary__f35d743d3b761113.srt	Full SRT sidecar. Normalized text similarity to canonical JSON: 1.0000; only the JSON segmentation supplies the displayed timestamps.

Historical question-source excerpts are shown only when a page has a strong text-based unit match. They are source references, not new solutions or exam predictions.

Known gaps, conflicts and corrections

1. 2 rendered pages in 001_UE26ME142A_Unit_1_Class_1.0_Lecture_PPT__a53eb8b08a5ac7ff.slides.md have no readable `.slides.md` text; their original page images are included.
2. 2 rendered pages in 001_UE26ME142A_Unit_1_Class_1.1_Lecture_PPT__a6f76a088910b54a.slides.md have no readable `.slides.md` text; their original page images are included.
3. 2 rendered pages in 002_UE26ME142A_Unit_1_Class_2.0_Lecture_PPT__de74ea957f21118c.slides.md have no readable `.slides.md` text; their original page images are included.
4. 2 rendered pages in 002_UE26ME142A_Unit_1_Class_2.1_Lecture_PPT__bb478549c01cfb60.slides.md have no readable `.slides.md` text; their original page images are included.
5. 2 rendered pages in 002_UE26ME142A_Unit_1_Class_2.2_Lecture_PPT__3fc8761d12196c6f.slides.md have no readable `.slides.md` text; their original page images are included.

6. 3 rendered pages in 003_UE26ME142A_Unit_1_Class_3_Lecture_PPT__40166a1710a03d41.slides.md have no readable `.slides.md` text; their original page images are included.
7. 3 rendered pages in 004_UE26ME142A_Unit_1_Class_4_Lecture_PPT__984b7c368532af06.slides.md have no readable `.slides.md` text; their original page images are included.
8. 3 rendered pages in 005_UE26ME142A_Unit_1_Class_5_Lecture_PPT__f49d361d79853022.slides.md have no readable `.slides.md` text; their original page images are included.
9. 2 rendered pages in 006_UE26ME142A_Unit_1_Class_6_Lecture_PPT__f79618b4a0f260c2.slides.md have no readable `.slides.md` text; their original page images are included.
10. 3 rendered pages in 008_UE26ME142A_Unit_1_Class_8_Lecture_PPT__2659a840cc2bfd13.slides.md have no readable `.slides.md` text; their original page images are included.
11. 2 rendered pages in 009_UE26ME142A_Unit_1_Class_9_Lecture_PPT__832f697eaf1d383c.slides.md have no readable `.slides.md` text; their original page images are included.
12. 2 rendered pages in 010_UE26ME142A_Unit_1_Class_10.0_Lecture_PPT__aa8f331eb00d810f.slides.md have no readable `.slides.md` text; their original page images are included.
13. 2 rendered pages in 010_UE26ME142A_Unit_1_Class_10.1_Lecture_PPT__fe73ddc3ca596069.slides.md have no readable `.slides.md` text; their original page images are included.
14. 2 rendered pages in 011_UE26ME142A_Unit_1_Class_11_Lecture_PPT__3624142116c7273d.slides.md have no readable `.slides.md` text; their original page images are included.
15. 2 rendered pages in 012_UE26ME142A_Unit_1_Class_12.0_Lecture_PPT__2e7b61b526d51f60.slides.md have no readable `.slides.md` text; their original page images are included.
16. 4 rendered pages in 012_UE26ME142A_Unit_1_Class_12.1_Lecture_PPT__1d60317fe4af0cbe.slides.md have no readable `.slides.md` text; their original page images are included.
17. No historical archive question page met the unit-match candidate threshold; all subject-level and ambiguous pages remain in datasets/question-archive.json.
18. This is a visibly labelled source-preservation draft. Human page-level semantic and visual review is not complete; no unsupported subject summary or formula correction is claimed.
19. The 19.26 GiB MP4 files remain hosted on Hugging Face; the full video path/hash and transcript coverage are indexed. The full audio/video was not watched end-to-end.
20. Formula candidates are machine-extracted text with source locators; use the adjacent original slide page and the formula index before treating a candidate as an authoritative equation.

21. The study PDF contains edited, topic-organized lecture notes and textual slide-page takeaways. Full-page slide images remain in the source/visual index datasets, not embedded as the lesson. Human content, ASR, formula, and page-level visual review remains pending.

Coverage

Evidence is summarized by source to keep the reading copy legible. Full source-unit IDs, locators and decisions remain in pack.json.

Source	Unit counts	Example decisions
SRC00034	5 covered · 0 excluded · 0 gaps	t000000-000299: Timestamp interval t000000-000299; canonical segment IDs s0001 to s0019. Sidecar variants compare... t000300-000599: Timestamp interval t000300-000599; canonical segment IDs s0020 to s0037. Sidecar variants compare... t000600-000899: Timestamp interval t000600-000899; canonical segment IDs s0038 to s0053. Sidecar variants compare...
SRC00035	5 covered · 0 excluded · 0 gaps	t000000-000299: Timestamp interval t000000-000299; canonical segment IDs s0001 to s0019. Sidecar variants compare... t000300-000599: Timestamp interval t000300-000599; canonical segment IDs s0020 to s0037. Sidecar variants compare... t000600-000899: Timestamp interval t000600-000899; canonical segment IDs s0038 to s0053. Sidecar variants compare...

Source	Unit counts	Example decisions
SRC00036	5 covered · 0 excluded · 0 gaps	t000000-000299: Timestamp interval t000000-000299; canonical segment IDs s0001 to s0019. Sidecar variants compare... t000300-000599: Timestamp interval t000300-000599; canonical segment IDs s0020 to s0037. Sidecar variants compare... t000600-000899: Timestamp interval t000600-000899; canonical segment IDs s0038 to s0053. Sidecar variants compare...
SRC00037	5 covered · 0 excluded · 0 gaps	t000000-000299: Timestamp interval t000000-000299; canonical segment IDs s0001 to s0019. Sidecar variants compare... t000300-000599: Timestamp interval t000300-000599; canonical segment IDs s0020 to s0037. Sidecar variants compare... t000600-000899: Timestamp interval t000600-000899; canonical segment IDs s0038 to s0053. Sidecar variants compare...

C Revision and retrieval practice